

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                          |
|----------------------|--------------------------|
| Product Name         | <u>Copper powder</u>     |
| CAS No               | 7440-50-8                |
| Synonyms             | Bronze Powder            |
| Molecular Formula    | Cu                       |
| Molecular Weight     | 63.54                    |
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | 42689                                                                                       |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                  |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | ANZinfo@thermofisher.com                                                                    |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number      HSR002948

### GHS Classification

#### Physical hazards

Flammable solids

Category 2

#### Health hazards

Based on available data, the classification criteria are not met

#### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 2

### Label Elements

**Signal Word****Warning****Hazard Statements**

H228 - Flammable solid

H400 - Very toxic to aquatic life

H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P370 + P378 - In case of fire: Use fire-fighting equipment on basis class D to extinguish

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

May form explosible dust-air mixture if dispersed

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

May form combustible dust concentrations in air

## Section 3 - Composition and Information on Ingredients

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Copper    | 7440-50-8 | <=100    |

## Section 4 - First Aid Measures

**Description of first aid measures****General Advice**

If symptoms persist, call a physician.

**New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

Dry sand, approved class D extinguishers.

### **Extinguishing media which must not be used for safety reasons**

Water may be ineffective.

### **Specific Hazards Arising from the Chemical**

Flammable. Dust can form an explosive mixture with air. Fine dust dispersed in air may ignite. Keep product and empty container away from heat and sources of ignition. Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Copper oxides.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Use personal protective equipment as required. Avoid dust formation. Ensure adequate ventilation.

#### **Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

#### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Prevent product from entering drains.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

### **Precautions for Safe Handling**

#### **Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Ensure adequate ventilation.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

#### Conditions for Safe Storage, Including any Incompatibilities

##### **Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame.

##### **Incompatible Materials**

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

#### Control parameters

##### **Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component | New Zealand WEL             | Australia                                              | ACGIH TLV                  | The United Kingdom                                                                                                                         |
|-----------|-----------------------------|--------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Copper    | TWA: 0.01 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr |

##### **Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Appropriate engineering controls

##### **Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Individual protection measures, such as personal protective equipment

##### **Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

##### **Hand Protection**

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                                           |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent) |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                                             |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                                                  |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                                                                      |                                          |
|------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid                                                                                |                                          |
| <b>Appearance</b>                              | Red brown                                                                            |                                          |
| <b>Odor</b>                                    | No information available                                                             |                                          |
| <b>Odor Threshold</b>                          | No data available                                                                    |                                          |
| <b>pH</b>                                      | No information available                                                             |                                          |
| <b>Melting Point/Range</b>                     | 1083 °C / 1981.4 °F                                                                  |                                          |
| <b>Softening Point</b>                         | No data available                                                                    |                                          |
| <b>Boiling Point/Range</b>                     | 2580 °C / 4676 °F                                                                    | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Not applicable                                                                       | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available                                                             |                                          |
| <b>Explosion Limits</b>                        | No data available                                                                    |                                          |
| <b>Flash Point</b>                             | No information available                                                             | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available                                                                    |                                          |
| <b>Decomposition Temperature</b>               | No data available                                                                    |                                          |
| <b>Viscosity</b>                               | Not applicable                                                                       | Solid                                    |
| <b>Water Solubility</b>                        | Insoluble                                                                            |                                          |
| <b>Solubility in other solvents</b>            | No information available                                                             |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                                                      |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 1628 °C                                                                     |                                          |
| <b>Density / Specific Gravity</b>              | 8.9200                                                                               |                                          |
| <b>Bulk Density</b>                            | No data available                                                                    |                                          |
| <b>Vapor Density</b>                           | Not applicable                                                                       | Solid                                    |
| <b>Particle characteristics</b>                | No data available                                                                    |                                          |
| <b>Other information</b>                       |                                                                                      |                                          |
| <b>Molecular Formula</b>                       | Cu                                                                                   |                                          |
| <b>Molecular Weight</b>                        | 63.54                                                                                |                                          |
| <b>Flammable solids</b>                        | Burning rate or burning time = > 5 minutes and <= 10 minutes > 2.2 mm/s or < 45 secs |                                          |
|                                                | Wetted zone passed - No                                                              |                                          |
| <b>Evaporation Rate</b>                        | Not applicable - Solid                                                               |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Moisture sensitive. Air sensitive.         |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |

|                                         |                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                     |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                                                |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Avoid dust formation, Keep away from open flames, hot surfaces and sources of ignition, Exposure to air, Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                                                                                                                                                     |
| <b>Hazardous Decomposition Products</b> | Copper oxides.                                                                                                                                                               |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

##### Product Information

|                   |                                    |
|-------------------|------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure. |
| <b>Eyes</b>       | Avoid contact with eyes.           |
| <b>Skin</b>       | Avoid contact with skin.           |
| <b>Ingestion</b>  | May be harmful if swallowed.       |

#### Numerical measures of toxicity

|                            |                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                                                            |
| <b>Oral</b>                | Based on available data, the classification criteria are not met<br>Data from closely analogous substances |
| <b>Dermal</b>              | Based on available data, the classification criteria are not met<br>Data from closely analogous substances |
| <b>Inhalation</b>          | Based on available data, the classification criteria are not met                                           |

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation              |
|-----------|-----------|-------------|------------------------------|
| Copper    |           |             | LC50 > 5.11 mg/L ( Rat ) 4 h |

|                                       |                                                                  |
|---------------------------------------|------------------------------------------------------------------|
| <b>(b) skin corrosion/irritation;</b> | Based on available data, the classification criteria are not met |
|---------------------------------------|------------------------------------------------------------------|

|                                           |                                                                  |
|-------------------------------------------|------------------------------------------------------------------|
| <b>(c) serious eye damage/irritation;</b> | Based on available data, the classification criteria are not met |
|-------------------------------------------|------------------------------------------------------------------|

|                                               |                                                                  |
|-----------------------------------------------|------------------------------------------------------------------|
| <b>(d) respiratory or skin sensitization;</b> |                                                                  |
| <b>Respiratory</b>                            | Based on available data, the classification criteria are not met |
| <b>Skin</b>                                   | Based on available data, the classification criteria are not met |

|                                    |                                                                  |
|------------------------------------|------------------------------------------------------------------|
| <b>(e) germ cell mutagenicity;</b> | Based on available data, the classification criteria are not met |
|------------------------------------|------------------------------------------------------------------|

|                             |                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>(f) carcinogenicity;</b> | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|

|                                   |                                                                  |
|-----------------------------------|------------------------------------------------------------------|
| <b>(g) reproductive toxicity;</b> | Based on available data, the classification criteria are not met |
|-----------------------------------|------------------------------------------------------------------|

|                                  |                                                                  |
|----------------------------------|------------------------------------------------------------------|
| <b>(h) STOT-single exposure;</b> | Based on available data, the classification criteria are not met |
|----------------------------------|------------------------------------------------------------------|

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed**

No information available.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                      | Water Flea                                       | Freshwater Algae                                                 | Microtox |
|-----------|--------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|----------|
| Copper    | Onchorhynchys mykiss:<br>LC50=0.15 mg/L 96h<br>Cuprinus carpio:<br>LC50=0.8 mg/L 96h | EC50: = 0.03 mg/L, 48h<br>Static (Daphnia magna) | 0.0426-0.0535 mg/L<br>EC50 72 h<br>0.031-0.054 mg/L EC50<br>96 h |          |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence** Insoluble in water, May persist.

**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** May have some potential to bioaccumulate Product has a high potential to bioconcentrate

**Mobility** Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

|                               |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                                                |
| <b>Other Information</b>      | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains. |

## Section 14 - Transport Information

### NZS 5433:2020

|                                |                                 |
|--------------------------------|---------------------------------|
| <b>UN-No</b>                   | UN3089                          |
| <b>Proper Shipping Name</b>    | Metal powder, flammable, n.o.s. |
| <b>Technical Shipping Name</b> | Copper                          |
| <b>Hazard Class</b>            | 4.1                             |
| <b>Packing Group</b>           | III                             |

### IATA

|                                |                                 |
|--------------------------------|---------------------------------|
| <b>UN-No</b>                   | UN3089                          |
| <b>Proper Shipping Name</b>    | Metal powder, flammable, n.o.s. |
| <b>Technical Shipping Name</b> | Copper                          |
| <b>Hazard Class</b>            | 4.1                             |
| <b>Packing Group</b>           | III                             |

### IMDG/IMO

|                                |                                 |
|--------------------------------|---------------------------------|
| <b>UN-No</b>                   | UN3089                          |
| <b>Proper Shipping Name</b>    | Metal powder, flammable, n.o.s. |
| <b>Technical Shipping Name</b> | Copper                          |
| <b>Hazard Class</b>            | 4.1                             |
| <b>Packing Group</b>           | III                             |

| Component                    | IMDG Marine Pollutant                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------|
| Copper<br>7440-50-8 ( ≤100 ) | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Environmental hazards</b>                                                    | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO                |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002948 |
|-----------------------------|-----------|

### **National Regulations**

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below



| Component | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL)                    |
|-----------|------------------------------------|--------------------------------------|----------------------------------------|--------------------------------------------------------|
| Copper    |                                    |                                      |                                        | 0.0014 mg/L (Freshwater)<br>0.0013 mg/L (Marine water) |

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                                                                   |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Copper    |                                                                                           |                                                                                          | IMDG regulated marine pollutant (Listed in the index, listed under Copper metal powder) |

**Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Copper    | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Copper    | 7440-50-8 | X     | X    | 231-159-6 | -      | -   | KE-08896 | X     | X    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-----------|-----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Copper    | 7440-50-8 | X    | ACTIVE                                        | X   | -   | X     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

First aid for chemical exposure, including the use of eye wash and safety showers.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

### Revision Date

31-May-2024

### Revision Summary

New emergency telephone response service provider

### Disclaimer

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## End of Safety Data Sheet